



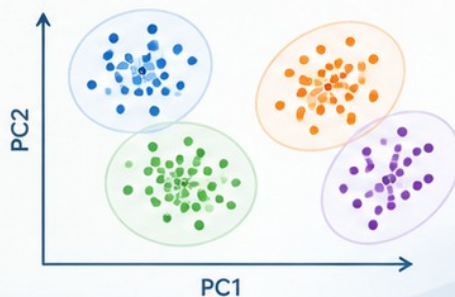
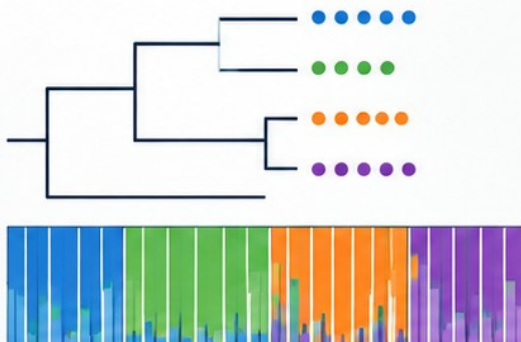
PR
STATS

Population Genomics and Population Structure Analysis

Nikolay Oskolkov, Group Leader (PI) at NIRI, Riga, Latvia



ATGCCTGAATCGTA
GTACCGTAACCTGA
TCGATGCCTGAATC
AACCTGATCGTACC
..





Metabolic Research Group at NIRI

NIRI NATIONAL
INSTITUTE
OF RESEARCH
AND INNOVATION

Latvian Institute of
Organic Synthesis



NEWS | ABOUT | TEAM | EVENTS | RESEARCH | DISSEMINATION | CONTACT US



Metabolic Research Group

Publications

Conferences

[Metabolic Research Group](#)

The Metabolic Research Group (MRG) focuses on advancing computational methods to identify and validate novel drug targets for metabolic diseases. Our research profile centers on the development and application of machine learning approaches, combined with statistical modeling, to extract biological knowledge from complex datasets. A key expertise of the group is the integration of diverse multiOmics data—including genomics, transcriptomics, proteomics, metabolomics, and metagenomics—enabling a systems-level understanding of metabolic processes and disease mechanisms. Through this integrative and data-driven approach, we aim to contribute to precision medicine by supporting the discovery of innovative therapeutic strategies within the TARGETWISE project.



Ph.D. Nikolay Oskolkov
Group Leader (PI) of the Metabolic
Research Group



Mohammad Banabazi, postdoc
in Metabolic Research Group



Kristina Grausa, PhD student
in Metabolic Research Group



Daniel Rivas, MD, postdoc
in Metabolic Research Group

1 more postdoctoral fellow and 1 PhD student to be hired

**If you know anyone who might be interested,
please contact me!**

About us

Organizer: Oliver Hooker, PR Stats courses

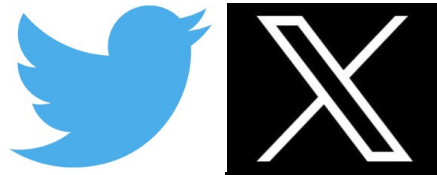
oliver@prstats.org



Instructor:

Dr. Nikolay Oskolkov, MRG Group Leader, NIRI, Riga,
lecturer at Lund University, Lund, Sweden

nikolay.oskolkov@niri.lv



@NikolayOskolkov



@oskolkov.bsky.social



Personal homepage:
<https://nikolay-oskolkov.com>



Brief introduction: who am I

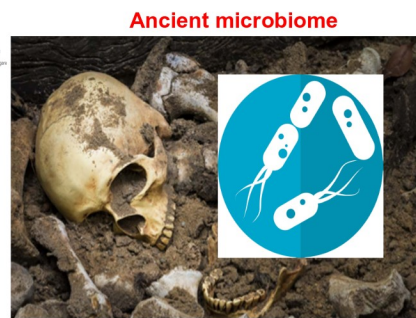
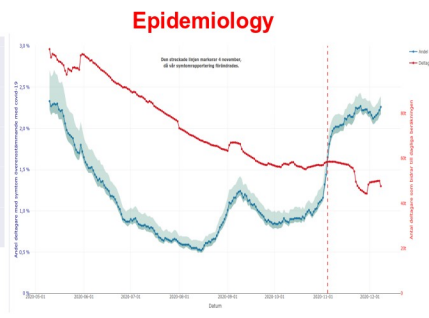
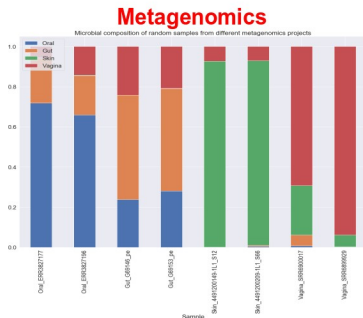
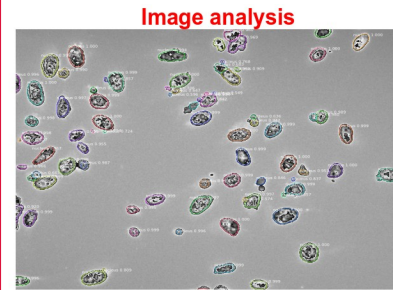
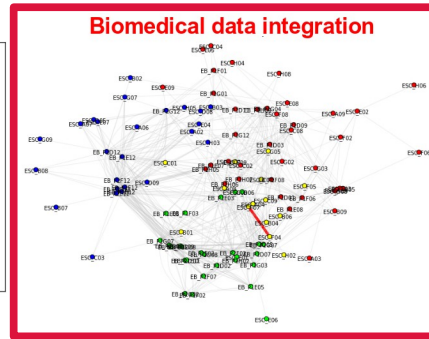
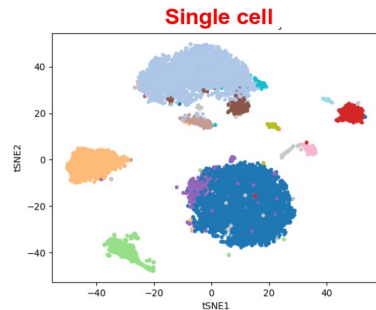
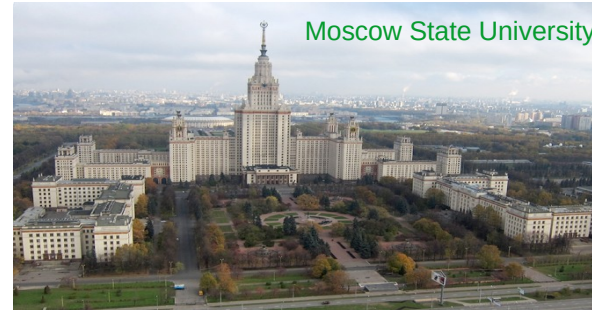
2007 PhD in theoretical physics in Moscow, Russia

2011 medical genomics at Lund University, Sweden

2016 bioinformatician at NBIS SciLifeLab, Sweden

2025 Metabolic Research Group leader, NIRI, Latvia

60+ publications; h-index = 28; 4,600+ citations



- Name
- University / Institute / Company
- Research interest(s)
- Previous experience with computational analysis and bioinformatics
- Motivation to join the course
- Expectations from the course

Day 1: introduction to next generation data analysis, GATK and ANGSD

Day2: dimensionality reduction with PCA, tSNE and UMAP

Day3: clustering methods, phylogenetic analysis, Admixture and Structure

Day4: selection scans with Fst, F-statistics for population structure, relation to PCA

Day5: capturing gene flow, introgression detection with D-statistic

Day 1

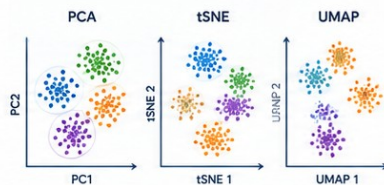
Next generation data analysis,
GATK and ANGSD



- NGS data and variant calling
- Best practices and quality control
- GATK workflows
- ANGSD for population genetic analysis

Day 2

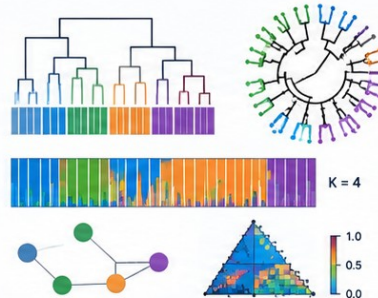
Dimensionality reduction
with PCA, tSNE and UMAP



- High-dimensional data
- PCA: linear structure
- tSNE: local structure
- UMAP: global structure

Day 3

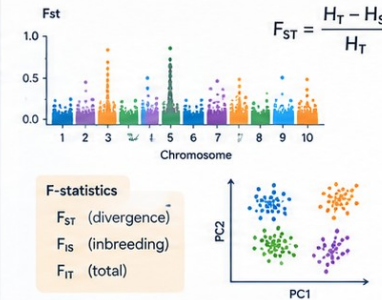
Clustering methods,
phylogenetic analysis,
Admixture and Structure



- Clustering (hierarchical, k-means, etc.)
- Phylogenetic trees
- Admixture analysis
- STRUCTURE / population structure

Day 4

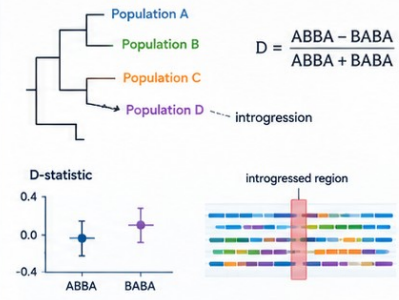
Selection scans with Fst,
F-statistics for population structure,
relation to PCA



- Fst and selection scans
- F-statistics (population structure)
- Relation to PCA

Day 5

Capturing gene flow,
introgression detection
with D-statistic



- Gene flow and introgression
- D-statistic (ABBA-BABA)
- Detecting introgressed regions
- Interpretation and biological context

Monday 28th – Classes from 15:00-21:00

Session 1- 02:30:00 – Introduction to genetic variation

Introduction to next generation sequencing, genetic variation, SNP discovery and variant calling with GATK workflow.

Break – 01:00:00

Session 2 – 02:30:00 – Genetic variation and low-coverage genomics

Genotype likelihoods, and analysis of low-coverage sequencing data using ANGSD, probabilistic genotype calling

Tuesday 29th – Classes from 15:00-21:00

Session 3 – 02:30:00 – Dimensionality reduction for population genomics I

Introduction to PCA, MDS, t-SNE, and UMAP; visualization of population structure; strengths, limitations, and practical applications.

Break – 01:00:00

Session 4 – 02:30:00 – Dimensionality reduction for population genomics II

The horseshoe effect, triangular PCA patterns, effects of uneven sampling, and why UMAP may be misleading in population genomics analyses.

Wednesday 30th – Classes from 15:00-21:00

Session 5 – 02:30:00 – Model-based clustering and ancestry inference

Introduction to clustering methods; ADMIXTURE and STRUCTURE; maximum likelihood and Bayesian approaches for ancestry estimation.

Break – 01:00:00

Session 6 – 02:30:00 – Ancestry inference from low-coverage data

NGSadmix and genotype likelihood-based approaches; ancestry inference without genotype calling; applications to ancient and modern populations; local ancestry inference and haplotype clustering using ChromoPainter and fineSTRUCTURE; detecting fine-scale population structure and demographic history.

Thursday 1st – Classes from 15:00-21:00

Session 7 – 02:30:00 – Population differentiation and selection scans

Fixation index (FST), genome-wide scans for selection, interpretation of population differentiation signals.

Break – 01:00:00

Session 8 – 02:30:00 – F-statistics and population history inference

F2 and F3 statistics; applications to the Allen Ancient DNA Resource (AADR); relationships between F-statistics, PCA, and phylogenetic trees.

Friday 2nd - Classes from 15:00-21:00

Session 9 – 02:30:00 – Detecting introgression with F4- and D-statistics

Theory and interpretation of F4-statistics, D-statistics, and the ABBA-BABA framework for detecting admixture and gene flow.

Break – 01:00:00

Session 10 – 02:30:00 – Case studies in human evolution and ancient DNA

- Please **ask your questions anytime**, it is ok to interrupt me, or you can raise your hand and ask. You can also **ask questions online in the zoom chat** or later by email.
- I kindly ask you to please keep your **webcams on for better communication**, this is of course not mandatory an up to you but will help me to adjust me explanations
- The course includes **10 lectures (~1 h each) and 10 practicals (~1.5 h each)**, there is a **15 min break** after each lecture and **1 h lunch break** every day of the course. We can potentially make longer breaks if needed.
- During practicals, we will use **Rmarkdown and Jupyter notebooks**. At the end of each practical, I will use rendered html-versions of the notebooks to go through command lines with my explanations.
- The material is based on data and problems from **computational biology**, however the concepts discussed are general and can be applied for other types of data.

The course will take place from 2 pm to 8 pm UK time (3 pm to 9 pm CET)

The course GitHub repository containing lectures and exercises is:

https://github.com/NikolayOskolkov/PopGen_PopStructure

Please bookmark this address!

PopGen_PopStructure

Public

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1 Branch

0 Tags

Go to file

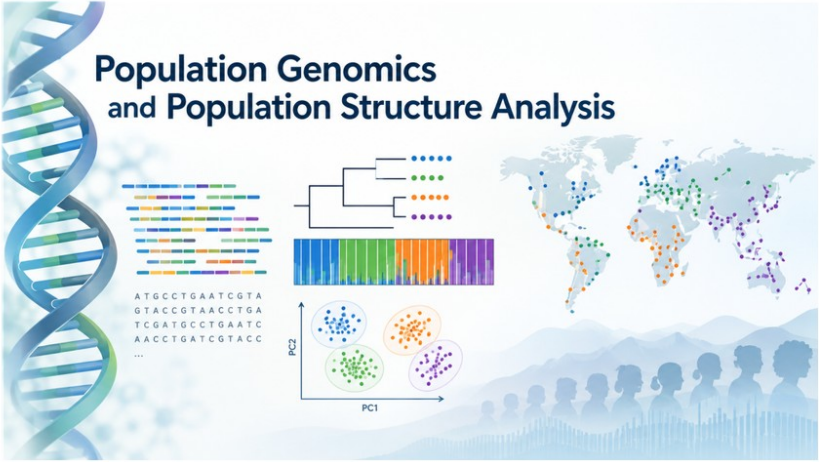
T

Add file

Code

NikolayOskolkov	edited README	8ec7121 · 13 minutes ago	5 Commits
articles	added articles	14 minutes ago	
images	added logo	32 minutes ago	
README.md	edited README	13 minutes ago	

README



The graphic features a blue DNA double helix on the left. To its right, a world map is populated with colored dots representing different populations. Below the map, a PCA plot shows three distinct clusters of points in blue, orange, and green, with axes labeled PC1 and PC2. Further left, a phylogenetic tree shows the evolutionary relationships between these clusters. At the bottom, silhouettes of a diverse group of people are shown against a light blue background.

Population Genomics and Population Structure Analysis

Instructor

- Dr. Nikolay Oskolkov, Group Leader (PI) at NIRI, Riga, Latvia

About

This repository contains material for population genomics course

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Contributors 1

NikolayOskolkov Nikolay Oskolkov

Google Colab

Python based



-  **Python**
(with rich ecosystem)
-  **Google Drive**
(easy file access)
-  **Free GPU/TPU**
(for ML and deep learning)
-  **No installation**
(runs in the browser)

VS

Post Cloud

RStudio and R based



-  **R**
(statistical computing)
-  **RStudio**
(full IDE)
-  **Dedicated servers**
(stable and secure)
-  **Large storage**
(for big datasets)

✓ Key strengths

- Easy to get started (no setup)
- Great for Python and machine learning
- Free access to GPUs/TPUs
- Seamless integration with Google Drive
- Ideal for notebooks and reproducible workflows

⚠ Limitations

- Python only (no RStudio)
- Session time limits (can disconnect)
- Less control over the computing environment
- Not ideal for large-scale or long-running jobs

✓ Key strengths

- RStudio with full IDE and packages
- Great for statistical analysis and visualization
- Stable, dedicated computing environment
- Suitable for large datasets and long-running analyses
- Supports collaborative and reproducible workflows

⚠ Limitations

- R only (no Python)
- No free GPU/TPU access (CPU only)
- Requires access to Post Cloud infrastructure
- Slightly more setup and learning curve for beginners



Both platforms support your research – choose the one that fits your needs

Google Colab for flexible, Python-based, cloud computing | Post Cloud for RStudio, R and dedicated resources

 Your Workspace

[+ New Space](#)

 Guide

! What's New

 Recipes

 Cheatsheets

[Help](#)

Current System Status

 Posit Community

Info

\$ Plans & Pricing

 [Terms and Conditions](#)

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

Untitled1*

```
1 gen <- t(matrix(c(1, 0, 2, 0, 2, 0, 2, 1, 1, 1, 0, 1, 0, 2, 1, 2, 1, 1, 1, 1, 0, 1, 0, 2, 0, 1
2 colnames(gen) <- paste0("Ind", 1:5))
3 rownames(gen) <- paste0("SNP", 1:7))
4 print(gen)
5
6 gen_centered <- scale(gen, center = TRUE, scale = FALSE) # mean centering
7 covariance <- t(gen_centered) %*% gen_centered # computing variance-covariance matrix
8 eig <- eigen(covariance) # performing eigen value decomposition of the variance-covariance matrix
9
10 barplot(eig$values / sum(eig$values), names = paste0("PC", 1:5))
11
```

11:1	(Top Level)
------	---

R Script ↕

Console Terminal × Background Jobs ×

R 4.6.1 · /cloud/project/ ↗
Type 'license()' or 'licence()' for distribution details.

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

Session restored from your saved work on 2026-Sep-27 14:27:56 UTC (3 hours ago)

```
gen <- t(matrix(c(1, 0, 2, 0, 2, 0, 2, 1, 1, 1, 0, 1, 0, 2, 1, 2, 1, 1, 1, 1, 1, 0, 1, 0, 2, 0, 1, 1, 0, 2,
1, 2, 0, 1, 0), 5, by = TRUE))
> colnames(gen) <- paste0("Ind", 1:5)
> rownames(gen) <- paste0("SNP", 1:7)
> print(gen)
```

Ind1 Ind2 Ind3 Ind4 Ind5

SNP1	1	1	1	0	0
SNP2	0	1	2	1	2
SNP3	2	1	1	0	1
SNP4	0	0	1	2	2
SNP5	2	1	1	0	0
SNP6	0	0	1	1	1
SNP7	2	2	1	1	0

```
> gen_centered <- scale(gen, center = TRUE, scale = FALSE) # mean centering
> covariance <- t(gen_centered) %*% gen_centered # computing variance-covariance matrix
> eig <- eigen(covariance) # performing eigen value decomposition of the variance-covariance matrix
> barplot(eig$values / sum(eig$values), names = paste0("PC", 1:5))
```

Environment History Connections Tutorial

Import Dataset 143 MiB

R Global Environment

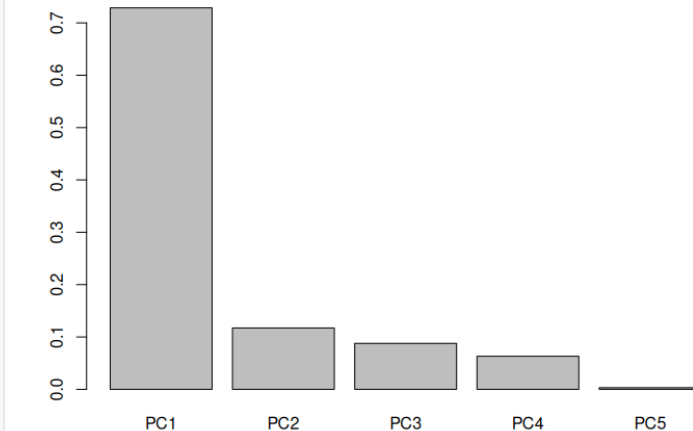
Category	Value
Category 1	Value 1
Category 2	Value 2
Category 3	Value 3
Category 4	Value 4
Category 5	Value 5
Category 6	Value 6
Category 7	Value 7
Category 8	Value 8
Category 9	Value 9
Category 10	Value 10
Category 11	Value 11
Category 12	Value 12
Category 13	Value 13
Category 14	Value 14
Category 15	Value 15
Category 16	Value 16
Category 17	Value 17
Category 18	Value 18
Category 19	Value 19
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Category 93	Value 93
Category 94	Value 94
Category 95	Value 95
Category 96	Value 96
Category 97	Value 97
Category 98	Value 98
Category 99	Value 99
Category 100	Value 100

covariance	num [1:5, 1:5] 6 3 -1 -3 -4 ...
------------	---------------------------------

eig	List of 2
-----	-----------

gen num [1:7, 1:5] 1 0 2 0 2 0 2 1 1 1

gen	non [1:7, 1:5] 1 0 2 0 2 0 2 1
gen_centered	num [1:7, 1:5] 0 -1 1 -1 1 ...



To run Jupyter notebooks on Google Colab please do the following:

1. Clone this repo: https://github.com/NikolayOskolkov/PopGen_PopStructure
2. Upload the repo to your Google drive
3. Start the Jupyter notebook from the Google drive and mount your google drive:

#Mount Google Drive to Colab

```
from google.colab import drive  
drive.mount('/content/drive')
```

```
!ls -l /content/drive/MyDrive/
```

```
import os  
os.chdir("/content/drive/MyDrive/data")
```

N.B. you need a Google account
to be able to run Google Colab!



Colab is now part of Google AI Plans. Get upgraded compute power and a suite of new AI tools all in one bundle.

Explore the Plans X

ANGSD_Lab_GoogleColab.ipynb ☆ ☁

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Q Commands + Code + Text ▶ Run all ▼



Share



✓ RAM 16GB Disk 100GB



Probabilistic Variant Calling with ANGSD on Low-Coverage Data

Course: Population Genomics and Population Structure Analysis**Instructor:** Nikolay Oskolkov**Platform:** Google Colab

In this tutorial, we will:

1. Install **Conda**, **ANGSD**, and **pcangsd** in Google Colab.
2. Mount **Google Drive** to access the input BAM files.
3. Prepare a list of BAM files with corrected paths.
4. Run **ANGSD** to compute genotype likelihoods (the `.beagle.gz` file).
5. Run **pcangsd** to perform PCA on the genotype likelihoods.
6. Plot the PCA result in **R**.
7. Run **NGSadmix** to estimate admixture proportions.
8. Plot the admixture bar plot in **R**.

Input data: 435 low-coverage BAM files from the 1000 Genomes Project, located in your Google Drive at `PopGen_PopStructure/1000G/smallbams/`.

1. Install Conda in Google Colab

Google Colab does not ship with Conda by default. We use the `condacolab` package, which installs a lightweight Miniconda environment directly inside the notebook.

Important: After running the cell below, the Colab kernel will **restart automatically**. This is expected — just continue with the next cell once the restart finishes.

```
# Install Conda inside Google Colab
!pip install -q condacolab
import condacolab
condacolab.install()
```

Announcement

Terminal

```
/content# ll
total 16
drwxr-xr-x 1 root root 4096 Sep 16 13:26 ./
drwxr-xr-x 1 root root 4096 Sep 27 17:29 ../
drwxr-xr-x 4 root root 4096 Sep 16 13:26 .config/
drwxr-xr-x 1 root root 4096 Sep 16 13:26 sample_data/
/content# cd sample_data/
/content/sample_data# ll
total 55512
drwxr-xr-x 1 root root 4096 Sep 16 13:26 ./
drwxr-xr-x 1 root root 4096 Sep 16 13:26 ../
-rwxr-xr-x 1 root root 1697 Jan 1 2000 anscombe.json*
-rw-r--r-- 1 root root 301141 Sep 16 13:26 california_housing_test.csv
-rw-r--r-- 1 root root 1706430 Sep 16 13:26 california_housing_train.csv
-rw-r--r-- 1 root root 18289443 Sep 16 13:26 mnist_test.csv
-rw-r--r-- 1 root root 36523880 Sep 16 13:26 mnist_train_small.csv
-rwxr-xr-x 1 root root 962 Jan 1 2000 README.md*
/content/sample_data# head README.md
This directory includes a few sample datasets to get you started.

* `california_housing_data*.csv` is California housing data from the 1990
US
Census; more information is available at:
https://docs.google.com/document/d/e/2PACX-1vRhYtsvc5e0R2FWNCwaBiKL6suI
0rxJig8LcS8bmCbyYsayia_DvP00BLXZ4CAIQ5nLDD8kTaIDRwrn/pub

* `mnist*.csv` is a small sample of the
[MNIST database](https://en.wikipedia.org/wiki/MNIST_database), which i
s
described at: http://yann.lecun.com/exdb/mnist/

/content/sample_data# ll
total 55512
drwxr-xr-x 1 root root 4096 Sep 16 13:26 ./
drwxr-xr-x 1 root root 4096 Sep 16 13:26 ../
-rwxr-xr-x 1 root root 1697 Jan 1 2000 anscombe.json*
-rw-r--r-- 1 root root 301141 Sep 16 13:26 california_housing_test.csv
-rw-r--r-- 1 root root 1706430 Sep 16 13:26 california_housing_train.csv
-rw-r--r-- 1 root root 18289443 Sep 16 13:26 mnist_test.csv
-rw-r--r-- 1 root root 36523880 Sep 16 13:26 mnist_train_small.csv
-rwxr-xr-x 1 root root 962 Jan 1 2000 README.md*
/content/sample_data#
```

[0] 0: bash* "d020aca8653c" 17:34 27-Sep-26